

STA 35C: Statistical Data Science III

Lecture 21: Natural Splines & Smoothing Splines

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Announcements

Homework 6 is posted

- Please start early and post questions on Piazza as needed
- Submit your solutions in a **single PDF**; if you submit multiple times, only your latest submission will be graded

Office hours today

- 3:30–4:30 PM at MSB 4220

Agenda

Last time: Basis functions and regression splines

- Basis functions let us fit nonlinear functions using least squares.
- Regression splines are piecewise polynomials joined smoothly at knots.

Today: Controlling spline flexibility

- Quick visual recap: truncated power basis
- Natural splines: improving boundary behavior
- Choosing knots and degrees of freedom
- Smoothing splines: penalized RSS and smoothness

Key idea: Splines are flexible nonlinear regression tools, but we must control their complexity through knots, degrees of freedom, or smoothness penalties.

Recap: Basis functions & regression splines

Basis-function model: Create transformed features $b_1(X), \dots, b_K(X)$, then fit

$$Y \approx \beta_0 + \sum_{k=1}^K \beta_k b_k(X).$$

- The fitted function can be nonlinear in X , but the model is linear in the coefficients β_k , so least squares still applies
- Examples:
 - *Polynomials:* $b_1(X) = X$, $b_2(X) = X^2, \dots$
 - *Step functions:* $b_1(X) = I_{(c_1, c_2]}(X)$, $b_2(X) = I_{(c_2, c_3]}(X), \dots$

Regression splines:

- *Piecewise polynomials* of degree d , joined smoothly at knots (cutpoints)
- *Smoothness constraints:* for a degree- d spline, f and its first $d - 1$ derivatives are continuous at each knot
- Degrees of freedom: a degree- d spline with K knots has $(d + 1) + K$ parameters

Recap: Why splines instead of high-degree polynomials?

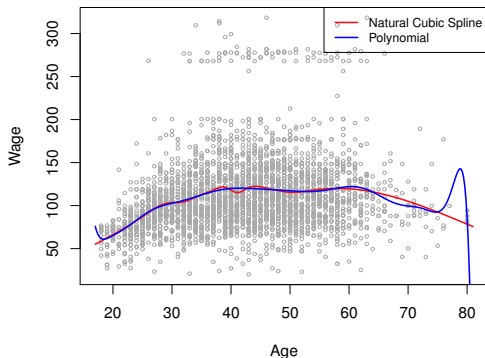


Figure: On the **Wage** data set, a natural cubic spline with 15 degrees of freedom (**blue**) vs. a degree-15 polynomial (**red**). Polynomials can oscillate excessively near the edges, while splines are more stable [JWHT21, Figure 7.7].

- Higher-degree polynomials can be flexible but often exhibit unwanted oscillations
- Splines keep the polynomial degree modest and add flexibility locally through *knots*, often giving more stable fits

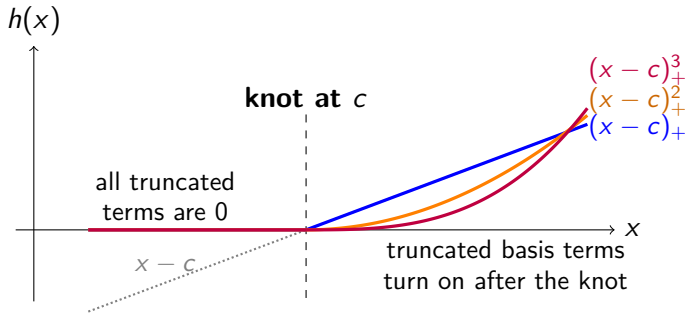
Recap: Truncated power basis

Truncated power basis for a degree- d spline:

$$\underbrace{1, X, X^2, \dots, X^d}_{\text{base polynomials}} \cup \left\{ \underbrace{(X - c_k)_+^d}_{\text{truncated power basis}} : k = 1 \dots K \right\}$$

where $(x - c)_+^d = \max\{x - c, 0\}^d$

- This representation **automatically enforces the desired smoothness** at each knot



Visual recap: What does $(x - c)_+$ do?

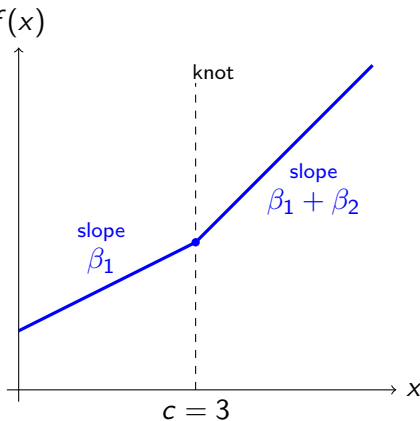
One-knot linear spline:

$$f(x) = \beta_0 + \beta_1 x + \beta_2 (x - 3)_+, \quad (u)_+ = \max(u, 0).$$

Equivalently,

$$f(x) = \begin{cases} \beta_0 + \beta_1 x, & x \leq 3, \\ (\beta_0 - 3\beta_2) + (\beta_1 + \beta_2)x, & x > 3. \end{cases}$$

- Before the knot: $(x - 3)_+ = 0$, so the term is inactive.
- After the knot: $(x - 3)_+$ turns on and changes the slope.
- The function remains continuous at $x = 3$.



(Optional) Continuity ensured by truncated power basis

A closer look at why/how truncated power basis ensures continuity:

- A function f is continuous at x_0 if

$$\lim_{x \rightarrow x_0 -} f(x) = f(x_0) = \lim_{x \rightarrow x_0 +} f(x)$$

- Observe that for $f(x) = (x - c_k)_+^d$,

$$\lim_{x \rightarrow c_k -} f(x) = 0,$$

$$f(c_k) = 0,$$

$$\lim_{x \rightarrow c_k +} f(x) = 0.$$

- You can similarly verify the continuity of derivatives; see Homework 6

Natural splines

Even a moderate-degree spline can exhibit wild curvature near the boundary

Natural splines¹ impose extra constraints at the boundary so that the spline is linear beyond the outermost knots:

- In practice, "natural spline" typically means a *natural cubic spline*
- For a natural cubic spline, the curve is constrained to be *linear beyond the boundary knots*; equivalently, the second derivative is 0 in the tails
 - Relative to a cubic spline with the same K interior knots, this reduces the degrees of freedom by 2
 - With K interior knots, a natural cubic spline has $K + 2$ degrees of freedom, compared with $K + 4$ for an ordinary cubic spline
- This usually restrains erratic tail behavior and yields narrower confidence intervals

¹A canonical basis for natural splines exists, but we skip details here. In R, see `splines::ns()`.

Illustration: Cubic spline vs. natural cubic spline

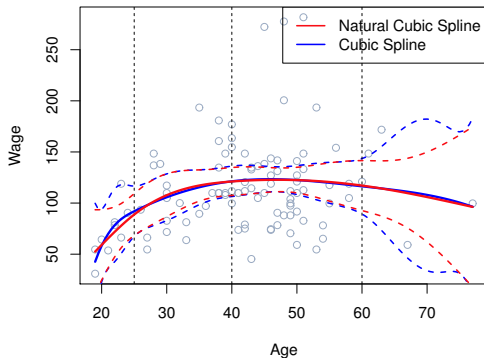


Figure: A cubic spline (blue) vs. a natural cubic spline (red) on a subset of the Wage data; vertical dashed lines show 3 knot locations. Note that natural spline is linear beyond the outer knots [JWHT21, Figure 7.4].

- Confidence intervals are narrower for natural splines
- Less risk of “wild” behavior near edges

Pop-up quiz: Natural splines

Suppose we compare an ordinary cubic spline and a natural cubic spline, both with K interior knots.

Question: Which statement is false?

- A) A cubic spline has $K + 4$ degrees of freedom.
- B) A natural cubic spline imposes extra boundary constraints to make the tails linear.
- C) The natural boundary constraints increase the degrees of freedom by 2.
- D) A natural cubic spline has $K + 2$ degrees of freedom.

Answer: C. The natural boundary constraints reduce, rather than increase, flexibility. Relative to a cubic spline with K interior knots, a natural cubic spline has 2 fewer degrees of freedom: $K + 2$ instead of $K + 4$.

Choosing knots and degrees of freedom

Knot placement choices:

- **Uniform width:** knots evenly spaced across the range of X .
- **Uniform mass:** knots placed at quantiles, so each region has roughly the same number of observations.
- **Domain knowledge:** place knots near meaningful breakpoints.

How much flexibility?

- More knots / higher degrees of freedom $\Rightarrow$ more flexible fit.
- Too little flexibility can underfit; too much can overfit.

Common practice:

- For smaller data sets, use a moderate number of knots (e.g. 3–5)
- For large data or highly nonlinear relationships, more knots might help
- Choose the number of degrees of freedom using cross-validation; knot placement is often chosen by quantiles or domain knowledge

How many knots should we have?

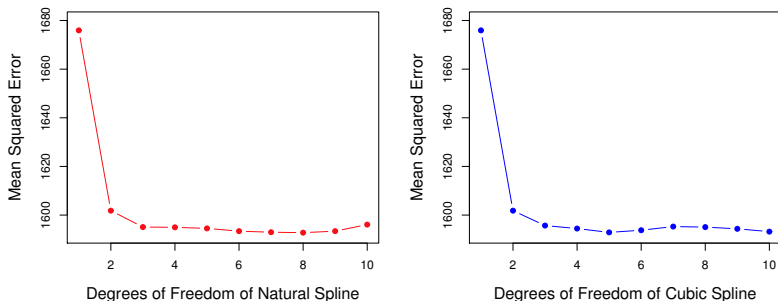


Figure: CV MSEs for different degrees of freedom in splines on the **Wage** data, modeling wage vs. age. **Left:** natural cubic spline. **Right:** cubic spline [JWHT21, Figure 7.6].

Question: How much flexibility should the spline have?

more knots / larger df $\iff$ more flexible fit

Common answer: choose degrees of freedom by cross-validation.

Example: Placing knots (uniform width vs. uniform mass)

Example

Setup: Suppose we have a hypothetical dataset with $n = 12$ points, $X \in [0, 12]$:

$$X = \{0.2, 0.9, 1.1, 2.2, 5.5, 6.0, 6.2, 8.0, 8.2, 10.8, 11.9, 12.0\}.$$

Two ways to pick knots (2 interior knots $\Rightarrow$ 3 segments):

1) **Uniform width:** Even spacing in $[0, 12]$.

- e.g. knots at $x = 4$ and $x = 8$.
- segments: $[0, 4]$, $(4, 8]$, $(8, 12]$.

2) **Uniform mass:** Each segment equally gets 4 data points

- After the 4th observation, place a knot near 2.2.
- After the 8th observation, place a knot near 8.0.
- segments: $[0, 2.2]$, $(2.2, 8]$, $(8, 12]$.

Pop-up quiz: Placing knots

Suppose X is highly unevenly distributed: many observations are near 0, and only a few are near 10. You want a spline with three intervals.

Question: Which statement is most accurate?

- A) Uniform-width knots always give each interval roughly the same amount of data.
- B) Quantile-based knots are designed to place roughly the same number of observations in each interval.
- C) More knots always improve test performance because the fit becomes more flexible.
- D) Knot placement does not matter once least squares is used.

Answer: B. Quantile-based knots adapt to the distribution of X , giving each interval roughly equal sample size. More knots can reduce bias but may increase variance, so flexibility should be controlled, often by cross-validation.

Smoothing splines: Formulation

Goal: Estimate a smooth function $g(x)$ that fits the data well without overfitting

We minimize a combination of (1) data fidelity and (2) a smoothness penalty:

$$\min_{g \in \mathcal{G}} \left\{ \underbrace{\sum_{i=1}^n (y_i - g(x_i))^2}_{\text{RSS}} + \lambda \underbrace{\int (g''(t))^2 dt}_{\text{smoothness penalty}} \right\}$$

- The parameter $\lambda \geq 0$ balances data fit vs. smoothness
 - $\lambda = 0$: prioritizes perfect fit to the training data, leading to (wiggly) interpolation
 - $\lambda \rightarrow \infty$: heavily penalizes curvature, so the solution approaches a least-squares line
- The solution turns out to be a *natural cubic spline* with knots at each of the observed x_i 's, but *shrunk* relative to a standard regression spline (due to penalty)

In R: The function `smooth.spline()` (in base R) performs smoothing spline fitting

Smoothing splines: Role of the curvature penalty

Why penalize $\int (g''(t))^2 dt$?

- The second derivative measures how sharply g bends
- A large $(g'')^2$ indicates “wavy” or erratic behavior
- Minimizing $\int (g'')^2$ forces smaller curvature, yielding a smoother shape

Interpretation of smoothness:

- The penalty $\int (g'')^2$ aims to control “roughness” or high-frequency wiggles
- A more wiggly g has bigger $(g'')^2$, thus a larger penalty

Regression splines vs. smoothing splines:

- **Regression splines:** fix knots/degree and enforce derivative continuity
- **Smoothing splines:** solve a *penalized* least squares problem; knots are placed at the observed x_i 's, but the penalty controls the effective flexibility.

Choosing the smoothing parameter λ

Effective degrees of freedom:

- As λ increases from 0 to ∞ , the solution transitions from an *interpolation* spline (exactly fitting all data) to a simple *straight line*
- The *effective* degrees of freedom, df_λ , correspondingly decreases from n to 2
 - Degrees of freedom = the number of free parameters
 - Although the solution has n knots, the penalty reduces the effective flexibility
- df_λ quantifies the spline's complexity, even though we do not explicitly choose knots

Selecting λ (or df_λ):

- Typically use cross-validation
 - Smoothing splines have a handy formula to compute LOOCV errors without re-fitting
- In practice:
 - Pick a small grid of λ -values (or df_λ -values)
 - Compute CV error and select the minimizer

If interested, see [JWHT21, Sec. 7.5.2] for technical details

Pop-up quiz: Smoothing splines

Suppose we increase the smoothing parameter λ in a smoothing spline.

Question: Which statement is most accurate?

- A) The curve becomes more wiggly, training RSS decreases, and effective degrees of freedom increase.
- B) The curve becomes smoother, training RSS generally increases, and effective degrees of freedom decrease.
- C) The curve becomes a step function because knots are removed one by one.
- D) The irreducible error decreases because the fitted curve is smoother.

Answer: B. Larger λ penalizes curvature more heavily, producing a smoother curve with lower effective degrees of freedom. The training RSS generally increases, but test performance may improve if variance is reduced.

R example: Fitting splines on a toy dataset

Data Setup:

```
set.seed(111)
x <- seq(0, 10, length.out=30)
y <- 2 + 3*sin(x) + rnorm(30, sd=0.3)
df <- data.frame(x, y)
```

Regression spline (cubic):

```
# Use 'bs()' from 'splines' package for B-spline basis
library(splines)

# Fit a cubic regression spline with, say, 2 internal knots
fit_spline <- lm(y ~ bs(x, degree=3, knots=c(3,7)), data=df)

# Predictions
x_new <- seq(0,10,length=200)
pred_spline <- predict(fit_spline, newdata=data.frame(x=x_new))
```

R example: Fitting splines on a toy dataset (cont'd)

Natural spline (cubic by default):

```
# 'ns()' builds a natural spline basis
fit_ns <- lm(y ~ ns(x, df=5), data=df)
pred_ns <- predict(fit_ns, newdata=data.frame(x=x_new))
```

Smoothing spline:

```
# 'smooth.spline()' solves the penalized objective
fit_smooth <- smooth.spline(x, y, df=6)
pred_smooth <- predict(fit_smooth, x=x_new)$y
```

Note: You can then plot $\hat{y}(x)$ for each model to compare

For more example codes, see [JWHT21, Sec. 7.8.1 & 7.8.2]

Wrap-up

- **Regression splines:**

- Piecewise polynomials + continuity constraints
- Truncated power basis for an easy linear-model fit
- “Natural” splines impose linear boundary conditions to avoid erratic tails
- Choose knot placement / number of knots via cross-validation

- **Smoothing splines:**

- A penalized approach balancing data fit vs. curvature
- The solution is a natural cubic spline with knots at each data point
- Effective degrees of freedom: from n to 2
 - $\lambda \rightarrow 0$ yields interpolation
 - $\lambda \rightarrow \infty$ yields a least-squares line
- Typically choose λ by cross-validation

References



Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani.

An Introduction to Statistical Learning: with Applications in R, volume 112 of *Springer Texts in Statistics*.

Springer, New York, NY, 2nd edition, 2021.